



a nanopore thin-film sensor in a reaction chamber, where immobilised primers on the nanofilm bind amplified LAMP products. Measurable signals are generated and can be analysed using a portable spectrometer, enabling point-of-care usage. The LAMP chip enables label-free pathogen detection with high sensitivity.

Quantum light breakthrough: Advancing technology and commercial applications

Researchers at the University of Copenhagen and Ruhr University Bochum have developed a way to control two quantum light sources rather than one. In replicating light sources, external disturbances pose challenges. However,

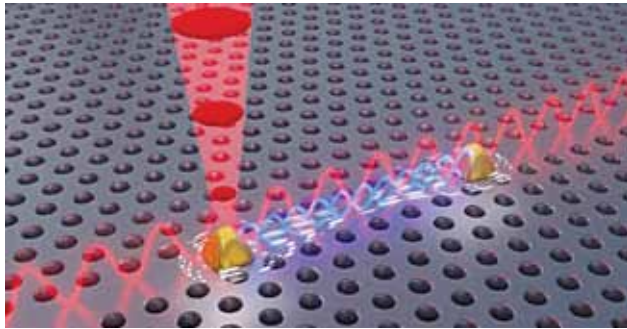


Illustration of two a chip comprising two entangled quantum light sources (Credit: NBI)

the team achieves two identical quantum light sources, a breakthrough in precise control. Quantum bits as 1 and 0 together offer immense processing power, surpassing supercomputers. 20-30 entangled sources can potentially create a universal error-corrected quantum computer. Transitioning to two quantum light sources requires silent nanochip development and meticulous control. The research sets the foundation for quantum physics research, while stakeholders can leverage it for advancing technologies like computers, the internet, and encryption.

Human trials of brain implant for dream control

Researchers from REMspace have conducted an experiment where they placed electrodes on the specific region of the brain's motor cortex that controls finger extensions in the left hand. The researchers have experimented by electrically stimulating the cortex during dream states. They implanted electrodes in a researcher's body, who also served as the subject. The subject was skilled in lucid dreaming, allowing observation of how finger stimulation influenced dreams. Notably, the stimulation did not wake the subject from their dream. The experiment revealed insights into

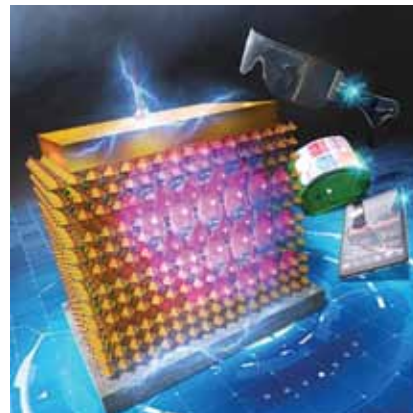
cortex stimulation for dream control, showing promise in inducing sensations, movements, and emotions. These signals can indicate the occurrence of lucid dreams.

A redox-based transistor as a reservoir system for neuromorphic computing

Researchers at Tokyo University of Science (TUS) and the National Institute for Materials Science (NIMS) in Japan have developed an ion-gating transistor that demonstrates enhanced reservoir states and short-term memory capabilities by utilising redox reactions. Ion-gating reservoirs control ion flow using electrodes and electrolytes. The voltage applied to the gate electrode triggers a reaction, modulating the drain current. Time-series data is converted into gate voltages, creating distinct reservoir states. A lithium-ion conducting glass ceramic (LICGC) is employed as the electrolyte. The researchers applied a 200nm LiCoO₂ film onto a 0.15mm LICGC substrate. The device outperforms memristors and spin torque devices in solving nonlinear equations with 40 reservoir states. Its nonlinearity, memory, and abundant reservoir states result in a low prediction error (0.163) in the NARMA2 task, which serves as a benchmark for complex time-series prediction.

Nanosheet technology for energy storage dielectric capacitors

Researchers at the Institute for Materials and Systems for Sustainability (IMaSS), Nagoya University in Japan, in collaboration with NIMS, have made a significant breakthrough in energy storage by developing a nanosheet device with unparalleled performance. Maximising the electric field of high dielectric constant is crucial for achieving superior energy density. The team utilised nanosheets in a perovskite structure, surpassing traditional studies. High



Ultrahigh energy storage in 2D high-K perovskites (Credit: Minoru Osada, Nagoya University)

polarisation dielectrics efficiently convert electric fields into energy, reaching record density levels. The nanosheet capacitors achieved much higher energy density (1-2 orders of magnitude higher than previous levels). They also demonstrated remarkable